

## lab 12

2026-04-23

```
install.packages("pcadapt")

## Installing package into '/home/t31d817/R/ifxrstudio/rstudio_3_20'
## (as 'lib' is unspecified)

## Warning in install.packages("pcadapt"): installation of package 'pcadapt' had
## non-zero exit status

install.packages("patchwork")

## Installing package into '/home/t31d817/R/ifxrstudio/rstudio_3_20'
## (as 'lib' is unspecified)

install.packages("viridis")

## Installing package into '/home/t31d817/R/ifxrstudio/rstudio_3_20'
## (as 'lib' is unspecified)

library(pcadapt)
library(vcfR)

##
##      *****      ***   vcfR   ***      *****
##      This is vcfR 1.15.0
##      browseVignettes('vcfR') # Documentation
##      citation('vcfR') # Citation
##      *****      *****      *****      *****

library(adegenet)

## Loading required package: ade4

##
##      /// adegenet 2.1.10 is loaded //////////////////////////////////
##
##      > overview: '?adegenet'
##      > tutorials/doc/questions: 'adegenetWeb()'
##      > bug reports/feature requests: adegenetIssues()

library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
```

```

##      intersect, setdiff, setequal, union
library(readr)
library(patchwork)
library(viridis)

## Loading required package: viridisLite
meta <- read_csv("/home/group/bioe-591-genomics/course-materials/data/selection/metadata.csv")

## Rows: 267 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (4): Sample.ID, Region, Region.ID, Site.ID
## dbl (2): Site.Latitude, Site.Longitude
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
x <- read_pcadapt("/home/group/bioe-591-genomics/course-materials/data/selection/salvelinus.vcf", type = "vcf")

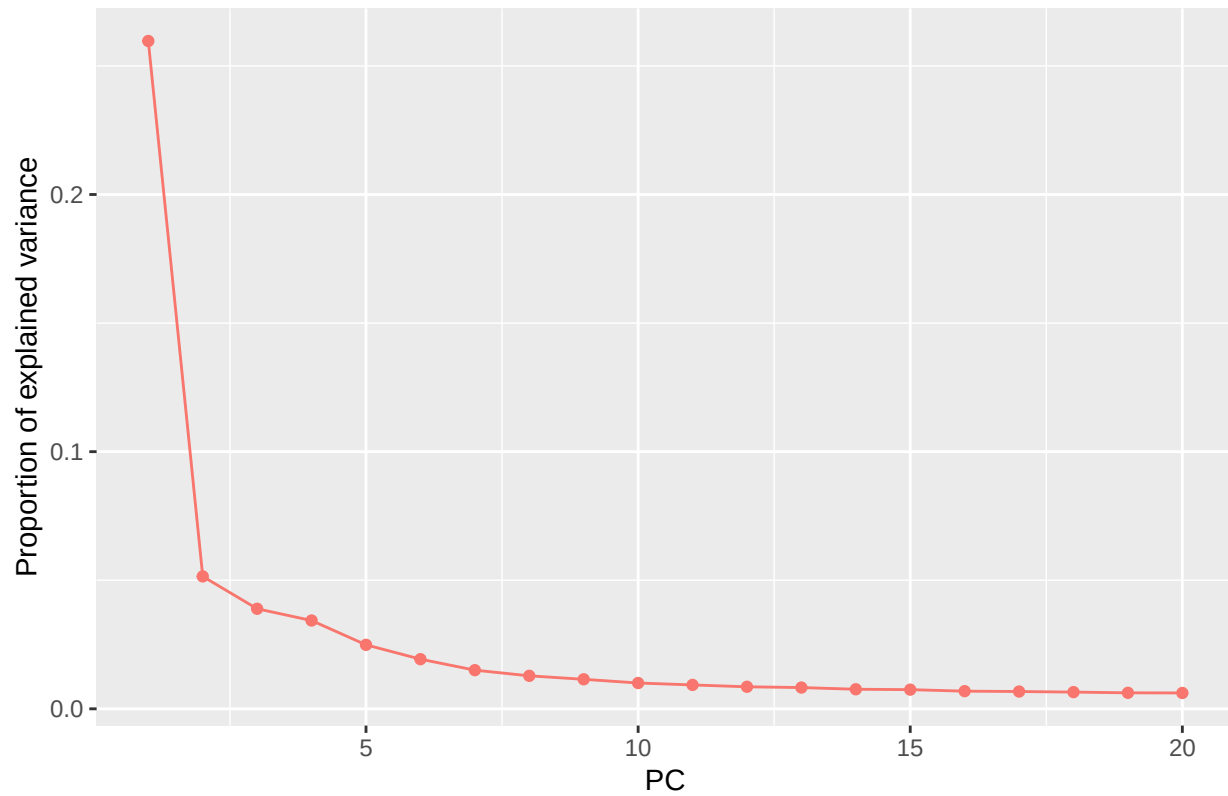
## Warning in file2other(input, type, match.arg(type.out), match.arg(allele.sep)):
## Converter vcf to pcadapt is deprecated. Please use PLINK for conversion to bed
## (and QC).

## No variant got discarded.
## Summary:
##
## - input file:          /home/group/bioe-591-genomics/course-materials/data/selection/salvelinus.vcf
## - output file:         /tmp/RtmpqCPHY4/file31dfe182b4e84.pcadapt
##
## - number of individuals detected: 267
## - number of loci detected:      24337
##
## 24337 lines detected.
## 267 columns detected.
pca <- pcadapt(x, K = 20)
plot(pca, option = "screplot")

## Warning: The '<scale>' argument of 'guides()' cannot be 'FALSE'. Use "none" instead as
## of ggplot2 3.3.4.
## i The deprecated feature was likely used in the pcadapt package.
## Please report the issue at <https://github.com/bcm-uga/pcadapt/issues>.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

Scree Plot – K = 20



```
pca_4 <- pcadapt(x, K = 4)
```

```
alpha <- 0.05  
padj <- p.adjust(pca_4$pvalues, method = "bonferroni")  
outlier_idx <- which(padj < alpha)  
neutral_idx <- which(padj >= alpha)  
cat("Outlier loci:", length(outlier_idx), "\n")
```

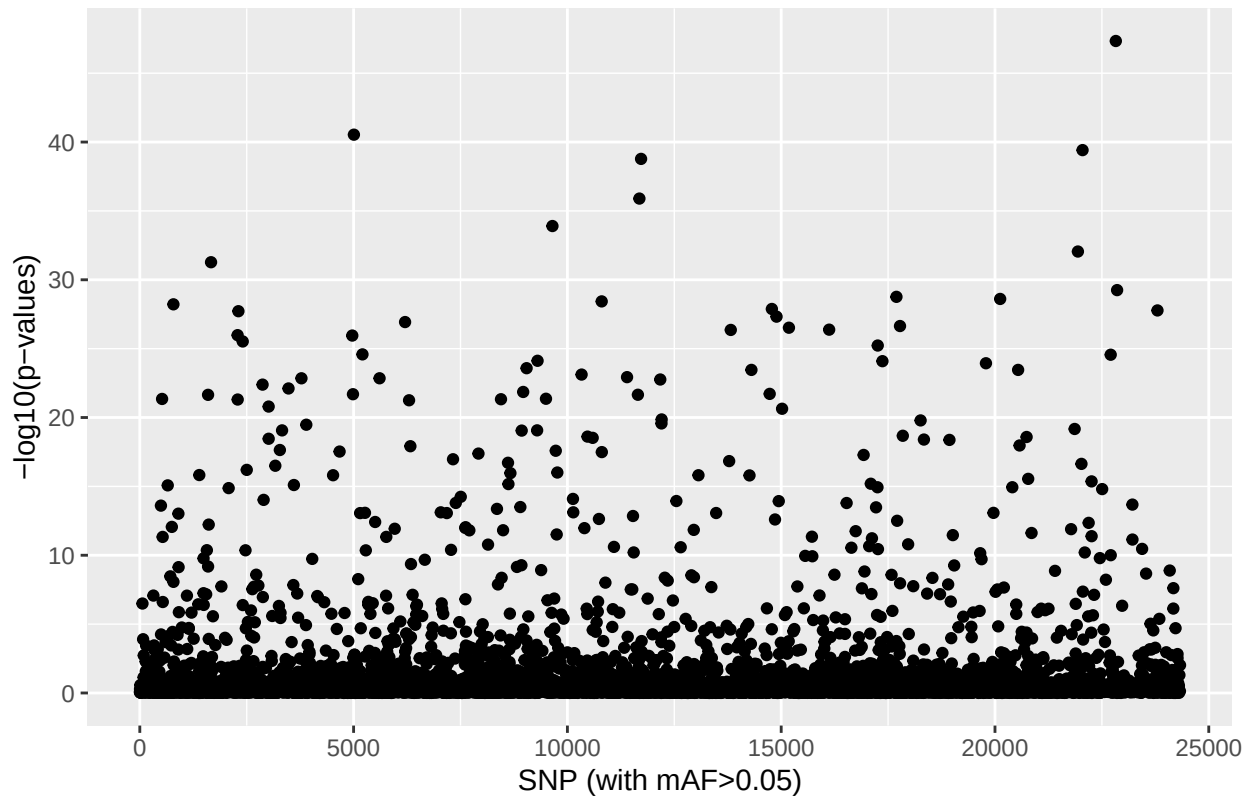
```
## Outlier loci: 306
```

```
cat("Neutral loci:", length(neutral_idx), "\n")
```

```
## Neutral loci: 5223
```

```
plot(pca_4)
```

## Manhattan Plot



```
vcf <- read.vcfR("/home/group/bioe-591-genomics/course-materials/data/selection/salvelinus.vcf", verbose = TRUE)
vcf_neutral <- vcf[neutral_idx, ]
vcf_outlier <- vcf[outlier_idx, ]
```

```
dna_neutral <- vcfr2DNABin(vcf_neutral, unphased_as_NA = FALSE, consensus = TRUE, extract.haps = FALSE)
```

```
## After extracting indels, 5223 variants remain.
```

```
gi_neutral <- DNABin2genind(dna_neutral)
dna_outlier <- vcfr2DNABin(vcf_outlier, unphased_as_NA = FALSE, consensus = TRUE, extract.haps = FALSE)
```

```
## After extracting indels, 306 variants remain.
```

```
gi_outlier <- DNABin2genind(dna_outlier)
```

```
gi_neutral_scaled <- scaleGen(gi_neutral, NA.method = "mean", scale = FALSE)
pca_neutral <- prcomp(gi_neutral_scaled, center = FALSE, scale. = FALSE)
gi_outlier_scaled <- scaleGen(gi_outlier, NA.method = "mean", scale = FALSE)
pca_outlier <- prcomp(gi_outlier_scaled, center = FALSE, scale. = FALSE)
```

```
# turn lists into dataframes
neutral_df <- data.frame(pca_neutral$x[, 1:2]) %>%
  mutate(sample = rownames(.))
outlier_df <- data.frame(pca_outlier$x[, 1:2]) %>%
  mutate(sample = rownames(.))
```

```
# join metadata
neutral_df <- left_join(neutral_df, meta, by = c("sample" = "Sample.ID"))
outlier_df <- left_join(outlier_df, meta, by = c("sample" = "Sample.ID"))
```

```

# shared axis limits
xlims <- range(c(outlier_df$PC1, neutral_df$PC1), na.rm = TRUE)
ylims <- range(c(outlier_df$PC2, neutral_df$PC2), na.rm = TRUE)

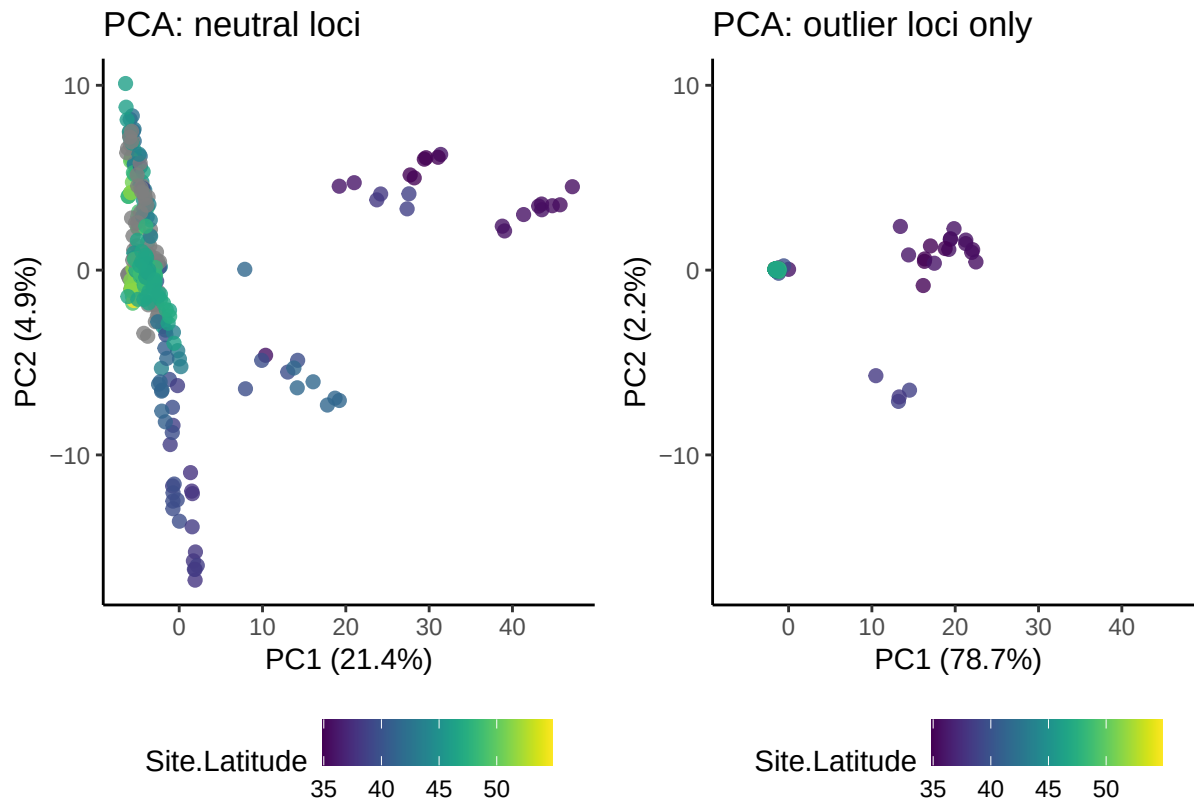
# percent variance explained
neutral_var <- 100 * summary(pca_neutral)$importance[2, ]
outlier_var <- 100 * summary(pca_outlier)$importance[2, ]

p1 <- ggplot(neutral_df, aes(x = PC1, y = PC2, color=Site.Latitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = xlims, ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: neutral loci",
    x = paste0("PC1 (", round(neutral_var[1], 1), "%)"),
    y = paste0("PC2 (", round(neutral_var[2], 1), "%)"),
  )

p2 <- ggplot(outlier_df, aes(x = PC1, y = PC2, color=Site.Latitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = xlims, ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: outlier loci only",
    x = paste0("PC1 (", round(outlier_var[1], 1), "%)"),
    y = paste0("PC2 (", round(outlier_var[2], 1), "%)"),
  )

p1 + p2

```



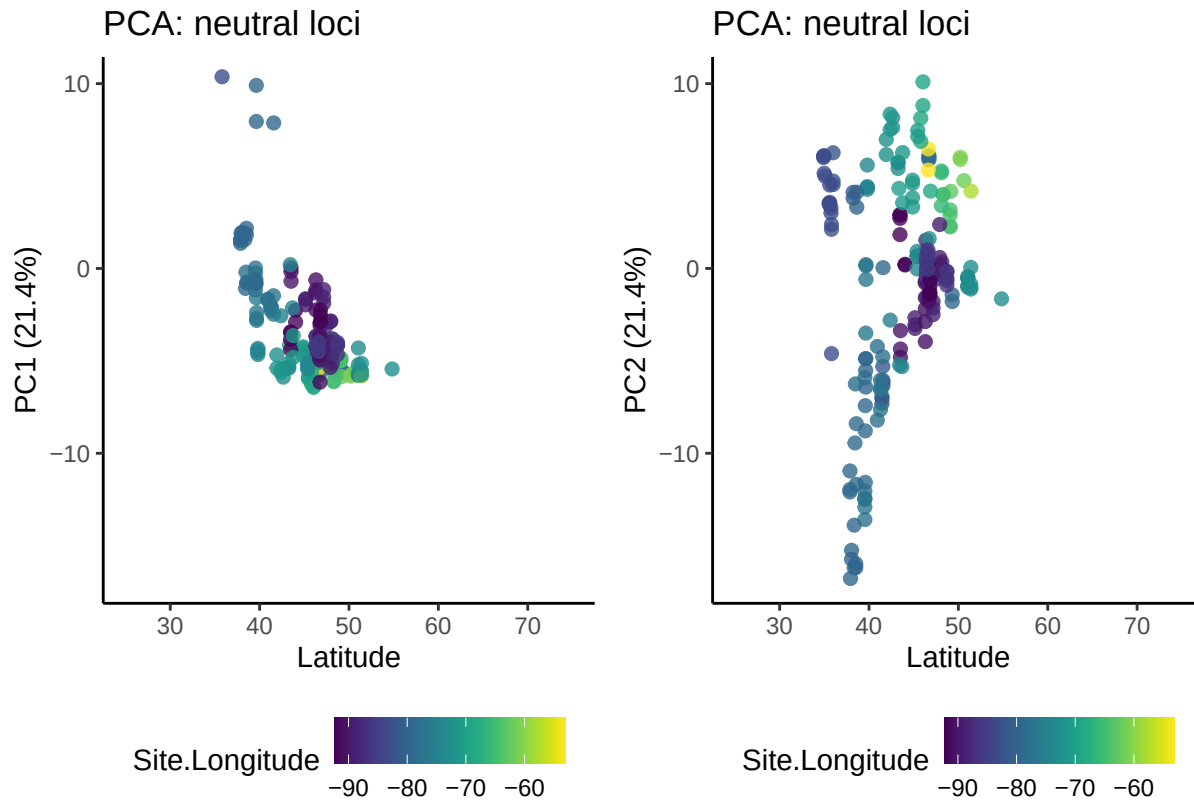
Only about 26% of the variability in the neutral loci data set is explained by this model but, it does look like there is some clustering based on site latitude. For the outlier loci significant more variability is explained by the model (~81%), the grouping within the PCA looks like it is in part due to site latitude.

```
p1latneu <- ggplot(neutral_df, aes(x = Site.Latitude , y = PC1, color= Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(25,75), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: neutral loci",
    y = paste0("PC1 (", round(neutral_var[1], 1), "%)"),
    x = paste0("Latitude")
  )

p2latneu <- ggplot(neutral_df, aes(x = Site.Latitude , y = PC2, color= Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(25,75), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: neutral loci",
    y = paste0("PC2 (", round(neutral_var[1], 1), "%)"),
    x = paste0("Latitude" )
  )
```

```
p1latneu + p2latneu
```

```
## Warning: Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
```

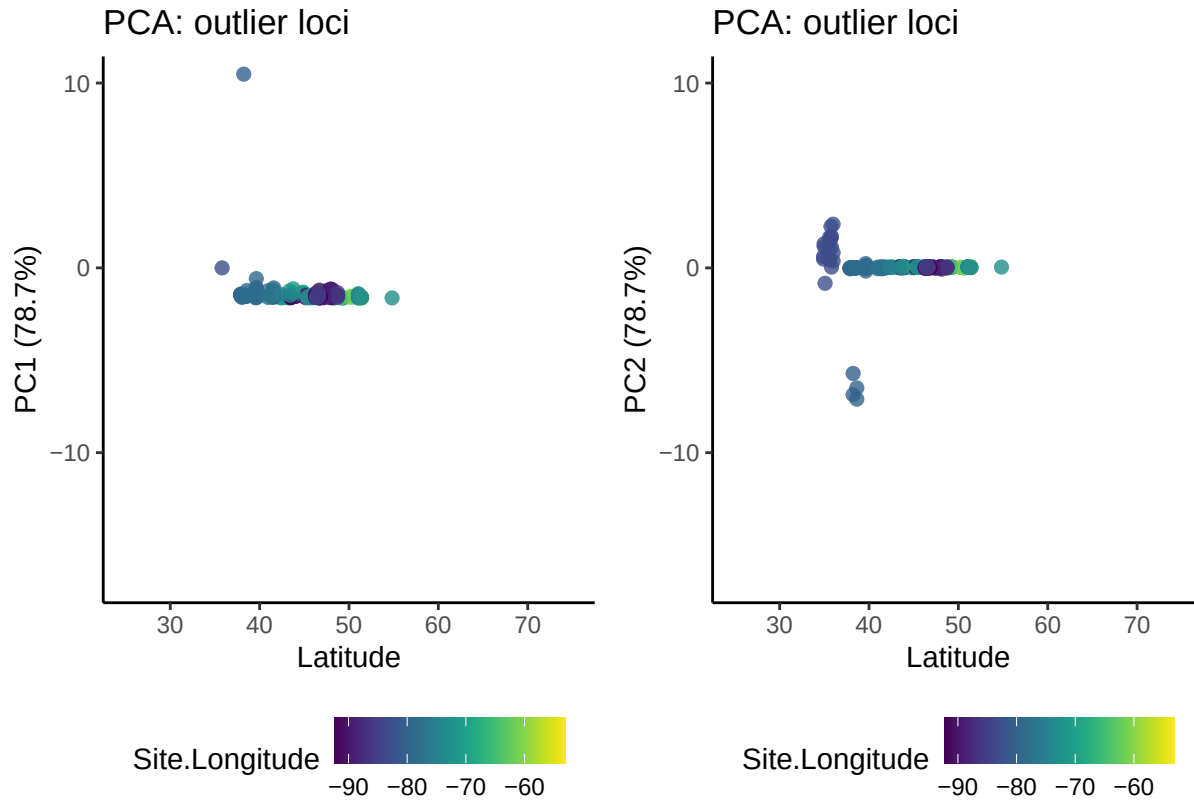


```
p1latout <- ggplot(outlier_df, aes(x = Site.Latitude , y = PC1, color= Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(25,75), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: outlier loci",
    y = paste0("PC1 (", round(outlier_var[1], 1), "%)",
    x = paste0("Latitude")
  )

p2latout <- ggplot(outlier_df, aes(x = Site.Latitude , y = PC2, color= Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(25,75), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: outlier loci",
    y = paste0("PC2 (", round(outlier_var[1], 1), "%)",
    x = paste0("Latitude" )
```

```
)
p1latout + p2latout
```

```
## Warning: Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Only about 20% of the variability in the data is captured with both PC1 and PC2 with latitude on the y axis. With PC1 there are two distinct clusters and with PC2 there are sections along the y axis with more grouping but there is a continuous group throughout. With the outlier loci about 78% of the variability was captured in the model. With PC1 all data except for an outlier is centered around -0.5 on the y axis and had an even spread across site latitude. With PC2 there is a cluster around the latitude 35 on the x-axis as well as another cluster below the rest of the data but the rest of the data is centered around 0 on the x-axis and evenly spread across latitudes.

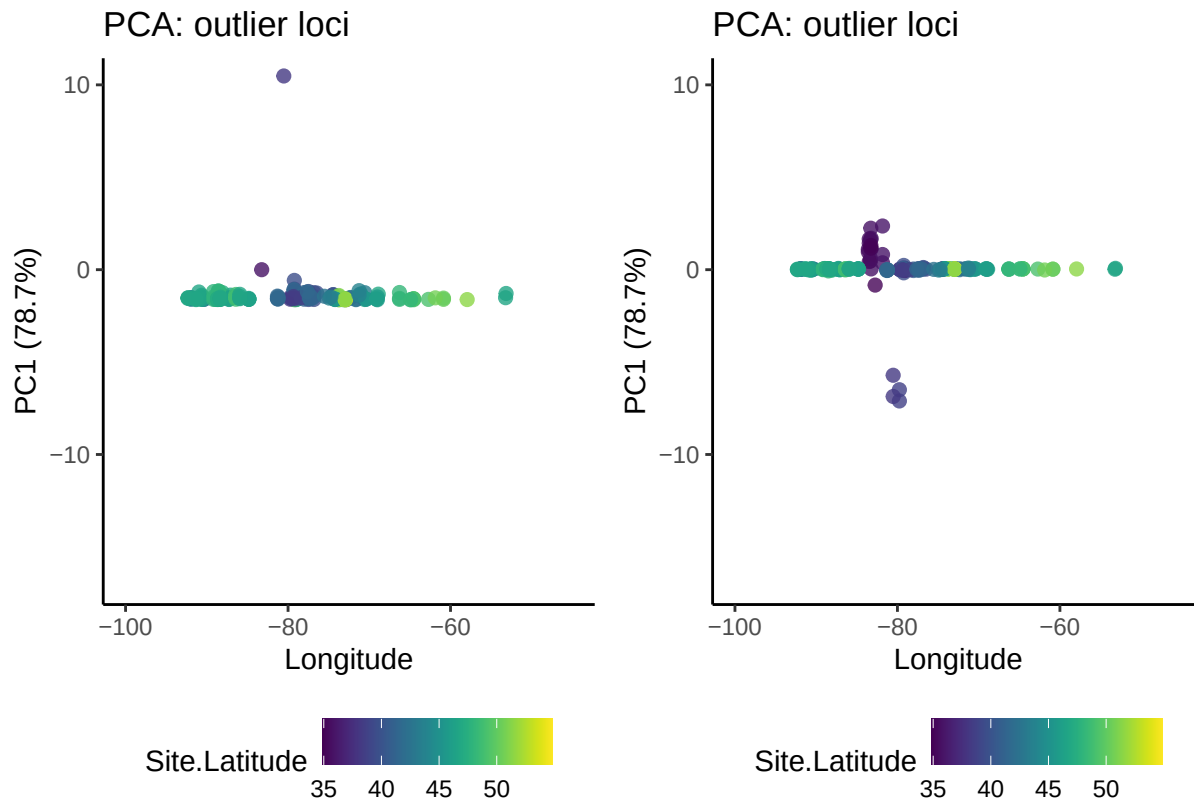
```
p1longout <- ggplot(outlier_df, aes(x = Site.Longitude , y = PC1, color= Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(-100,-45), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: outlier loci",
    y = paste0("PC1 (", round(outlier_var[1], 1), "%)",
    x = paste0("Longitude")
  )
p2longout <- ggplot(outlier_df, aes(x = Site.Longitude , y = PC2, color= Site.Longitude)) +
```



```
geom_point(size = 2, alpha = 0.8) +
scale_color_viridis() +
coord_cartesian(xlim = c(-100,-45), ylim = ylims) +
theme_classic() +
theme(legend.position = "bottom") +
labs(
  title = "PCA: outlier loci",
  y = paste0("PC1 (", round(outlier_var[1], 1), "%)"),
  x = paste0("Longitude")
)
```

p1longout + p2longout

```
## Warning: Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
p1longneu <- ggplot(neutral_df, aes(x = Site.Longitude , y = PC1, color= Site.Latitude)) +
geom_point(size = 2, alpha = 0.8) +
scale_color_viridis() +
coord_cartesian(xlim = c(-100,-45), ylim = ylims) +
theme_classic() +
theme(legend.position = "bottom") +
labs(
  title = "PCA: neutral loci",
  y = paste0("PC1 (", round(neutral_var[1], 1), "%)"),
  x = paste0("Longitude")
)
```

```

)
p2longneu <- ggplot(neutral_df, aes(x = Site.Longitude , y = PC2, color= Site.Latitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = c(-100,-45), ylim = ylims) +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(
    title = "PCA: neutral loci",
    y = paste0("PC1 (", round(neutral_var[1], 1), "%)",
    x = paste0("Longitude")
  )
)

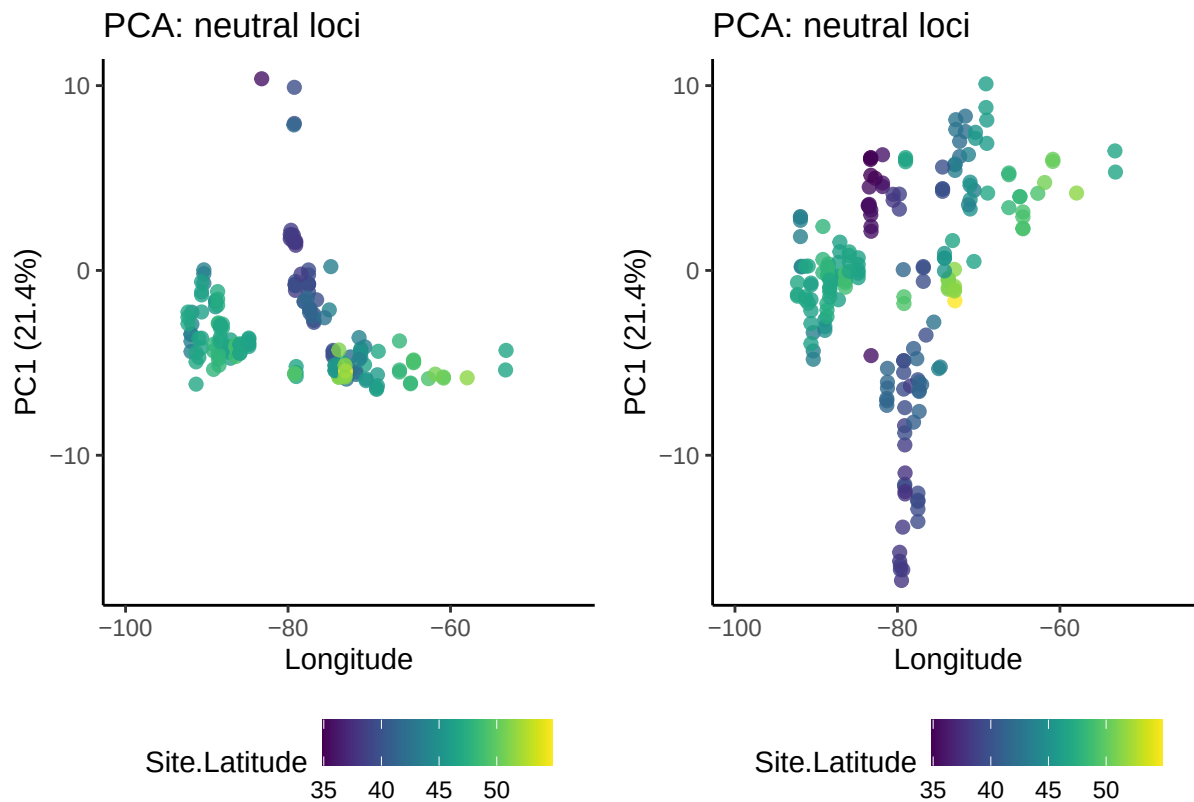
p1longneu + p2longneu

```

```

## Warning: Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 66 rows containing missing values or values outside the scale range
## ('geom_point()').

```



Outlier loci are evenly spread across the x-axis, in the PC1 plot there are a couple single outliers but in the PC2 plot there are two clusters that all have a latitude of around 36. In the plots of the neutral loci there are no real patterns following the longitude gradient some grouping appears for different site latitudes but the groups are not very distinct. PC1 has a more even spread across longitude but clustering still appears to be in part due to latitude.